

JYOTIRADITYASINH CHAUHAN

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Education

University of Houston

Bachelor of Science in Computer Science | GPA: 3.68

August 2023 – May 2027

Houston, TX

Experience

Cybersecurity Intern

May 2026 - August 2026

LyondellBasell

Houston, TX

- Piloted a SQL-backed (SQLite) vulnerability management pipeline at a live LyondellBasell site, automating ingestion of asset and vulnerability data via a Python poller pulling delta updates from Claroty CTD's REST API in near real-time.
- Engineered a CVSS v3.1 environmental recast engine that re-scores NVD base vulnerability ratings against OT-specific exploitability constraints (Adjacent attack vector, High attack complexity, High privileges required) and impact constraints (Low confidentiality requirement), correcting for IT-centric scoring assumptions that overstate real risk on segmented Process Control Network assets.
- Authored an end-to-end vulnerability management playbook covering a repeatable identify-prioritize-remediate process, Claroty CTD configuration requirements, and a four-table relational schema (assets, vulnerabilities, recast results, remediation status) feeding a BI dashboard, scaling the pilot toward all **45 sites**.

Research Assistant

Jan 2026 – Present

University of Houston

Houston, TX

- Built and shipped a full-stack web application with a GPU-hosted LLM backend integrated into Canvas LMS via Python/FastAPI; designed the end-to-end data pipeline — ingestion, inference (sglang), and response delivery — serving 100+ active users.
- Instrumented the system with structured logging and query analytics; performed systematic troubleshooting to iterate on pipeline performance and reduce latency, improving reliability under real production load.

AI/ML Intern

Sep 2025 – Dec 2025

Tata Consultancy Services

Houston, TX

- Accelerated POC delivery by 30% by building reusable Python, C#, and JavaScript frameworks alongside lightweight ETL pipelines, enabling faster data modeling and product experimentation across multiple client-facing engagements.
- Wrote complex queries and data transformation logic across large datasets to surface trends and drive recommendations; authored system diagrams and technical documentation to ship well-documented, extensible data systems.

Teaching Assistant

May 2024 – Aug 2025

University of Houston

Houston, TX

- Designed structured labs covering data structures, algorithms, operating systems, and computer architecture for 100+ students, improving debugging and troubleshooting efficiency by 25% through clear, repeatable workflows.
- Applied a data-informed approach to iteratively refine content each semester, leading live coding sessions and gathering feedback to continuously improve learning outcomes.

Projects

Embedded Systems Network Monitor | C++, Python, firmware interfaces, embedded systems, computer architecture Jan – Apr 2026

- Developed a lightweight C++ network monitoring agent targeting embedded systems with constrained resources; interfaced with firmware-level diagnostics to capture and analyze real-time telemetry across device nodes.
- Built a Python dashboard to visualize packet-level data and system health metrics, applying performance optimization techniques to reduce processing overhead by 30% on low-power hardware targets.

Large-Scale Log Analytics Pipeline | Splunk, SPL, Python, SQL, JavaScript, web application development

Oct – Dec 2025

- Built SPL correlation searches, interactive web application dashboards (JavaScript/D3.js), and alerting pipelines in Splunk to analyze large volumes of security event data; reduced simulated intrusion investigation time by 25%.
- Authored end-to-end data workflows from raw log ingestion to structured dashboards, standardizing evidence collection and translating complex multi-source datasets into clear, repeatable analytical procedures.

Extracurriculars

Information Security Lead

May 2023 – Present

CougarCS – Information Security Branch

- Designed and led hands-on workshops on network traffic analysis and troubleshooting using Wireshark for 30+ students, growing weekly attendance 25% by iterating on content structure based on participant feedback.
- Translated dense technical datasets and packet-level analysis into engaging, structured learning experiences — sharpening communication skills and a user-first approach to presenting complex data.

Skills

Languages: Python, C++, Java, JavaScript, TypeScript, SQL, R

Systems: Embedded systems, firmware interfaces, operating systems, computer architecture, distributed computing

Operational Technology: Claroty CTD, OT/ICS security, Purdue Model network segmentation, Process Control Networks, CVSS v3.1 environmental scoring, EPSS

Data & Visualization: ETL pipelines, PostgreSQL, SQLite, Tableau, Power BI, Splunk

Backend / Infra: FastAPI, Docker, Kubernetes, CI/CD, Git, Redis, REST APIs, JWT authentication, web application development

AI / ML: LLMs, RAG, inference optimization, sglang, benchmarking

Certifications: CompTIA Security+